Registry Run Key Persistence Detection and Investigation using Sysmon and Wazuh

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect Registry Run Key persistence techniques commonly associated with malware autorun behavior, user logon persistence, and long-term attacker access retention.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Registry Run Keys / Startup Folder	T1547.001
Execution	PowerShell	T1059.001

Attack Simulation

The following command was executed in the Windows endpoint lab environment:

```
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run
/v EvilPersistence /t REG_SZ /d notepad.exe /f
```

Attack Overview

Step	Activity
1	Registry Run Key persistence created
2	Sysmon captured registry modification
Step	Activity

3	Wazuh generated alert
4	System reboot / user logon occurred
5	Explorer.exe triggered autorun execution
6	Persistence payload executed automatically

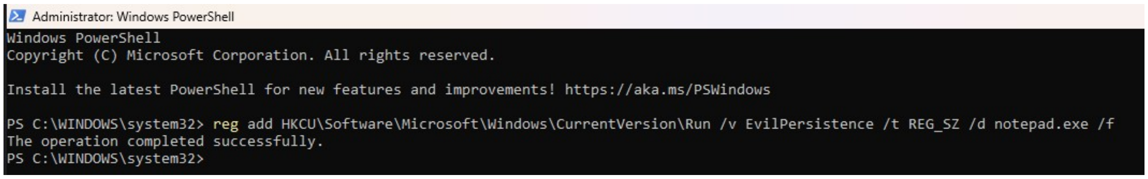

Fig-1Registry Run Key persistence simulation executed in the lab environment

Registry Persistence Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
13	Registry Value Modification
1	Process Creation
11	File Creation (related persistence artifact)

These events formed the primary telemetry chain used during the persistence investigation.

Sysmon Event ID 1 – Process Creation

Key Findings:

Field	**Value**
Event ID	1
Image	C:\Windows\System32\reg.exe
Parent Process	powershell.exe
User	socadmin
Integrity Level	High
Process ID	6544

Command Executed:

reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v EvilPersistence /t REG_SZ /d notepad.exe /f

Investigation Analysis:

Sysmon Event ID 1 recorded execution of the Windows Registry Console utility: reg.exe

The process was launched from: powershell.exe

indicating that PowerShell was used to establish persistence through Registry modification.

The command created a new Registry Run Key:

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

with:

Value Name: EvilPersistence

Data: notepad.exe

The /f parameter automatically overwrote any existing value without prompting the user.

This event is significant because it provides direct evidence of the persistence creation activity before the Registry itself was modified.

From an investigation perspective, Sysmon Event ID 1 reveals:

- The exact command executed
- The user responsible
- The parent process
- The persistence mechanism being created

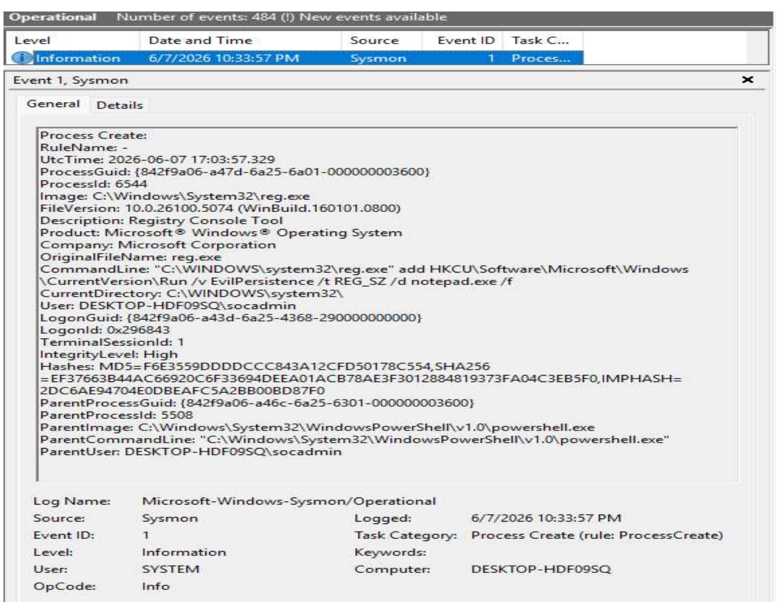

Fig: Sysmon Event ID 1 showing reg.exe creating a Registry Run Key persistence mechanism through PowerShell.

Sysmon Event ID 13 — Registry Persistence Creation Analysis

Key Findings

Field	Value
Event ID	13
Source Process	reg.exe
Field	Value
Registry Path	HKU SID \Software\Microsoft\Windows\CurrentVersion\Run
Registry Value	EvilPersistence
Payload	notepad.exe
User	socadmin

Critical Persistence Registry Path

```
HKCU\Software\Microsoft\Windows\CurrentVersion\Run
```

Sysmon internally translated the registry path into:

```
HKU\<SID>\Software\Microsoft\Windows\CurrentVersion\Run
```

This is expected behavior and is important for SOC analysts to recognize during investigations.

Investigation Analysis

Sysmon Event ID 13 captured the creation of a Registry Run Key persistence mechanism using `reg.exe`.

The Run Key persistence location is a high-value Windows autorun location because applications configured within this registry path automatically execute during user logon.

The persistence value configured during testing was:

```
notepad.exe
```

Although the payload itself was benign within the lab environment, the behavioral pattern remained highly suspicious because attackers frequently abuse Run Keys to maintain persistence after reboot or user logout.

Important SOC Analyst Note

The Sysmon configuration automatically mapped the registry persistence activity to:

```
T1060, RunKey
```

However, modern MITRE ATT&CK mapping more accurately aligns with:

Technique	ID
Registry Run Keys / Startup Folder	T1547.001

This highlights the importance of analyst validation instead of relying entirely on automated telemetry tagging.

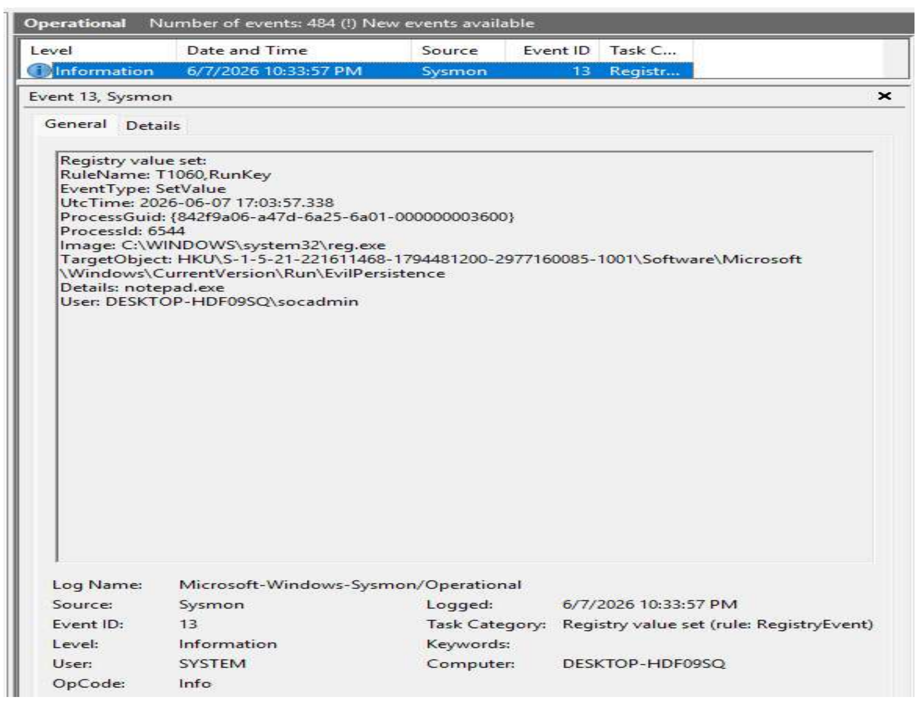

Fig- 2 Sysmon Event ID 13 showing Registry Run Key persistence creation

Wazuh Detection Analysis

During the persistence simulation, Wazuh generated multiple alerts that detected both the Registry Run Key creation activity and the associated command-line execution used to establish persistence.

The alerts were generated from correlated Sysmon Event ID 1 and Sysmon Event ID 13 telemetry collected from the monitored Windows endpoint.

Rule 92041 – Suspicious Registry Modification Activity

Field	Value
Rule ID	92041
Severity	10
Description	Value added to registry key has Base64-like pattern
Process	reg.exe
Parent Process	powershell.exe
User	socadmin
Command	reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run ...
MITRE	T1027, T1112

Analysis

Rule 92041 was triggered when:

reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v EvilPersistence /t REG_SZ /d notepad.exe /f

was executed through:

powershell.exe
↓
reg.exe

Wazuh correlated the process creation event and identified suspicious registry modification behavior associated with persistence establishment.

The alert provided visibility into the relationship between PowerShell execution and Registry modification activity.

Rule 92302 – Registry Run Key Persistence Detection

Field	Value
Rule ID	92302
Severity	6
Description	Registry entry to be executed on next logon was modified using command line application reg.exe
Source Process	C:\Windows\System32\reg.exe
Registry Path	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\EvilPersistence
Registry Value	notepad.exe
Event	SetValue

Field	Value
Type	
User	socadmin
Sysmon Event	Event ID 13
MITRE	T1547.001

Analysis

Rule 92302 was generated after Sysmon Event ID 13 recorded a Registry Run Key modification.

Registry Path: HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Registry Value: EvilPersistence

Data: notepad.exe

This alert confirmed creation of a persistence mechanism that would automatically execute after user logon.

The detection directly mapped to: T1547.001
Registry Run Keys / Startup Folder

which is a common persistence technique used by malware and threat actors.

Fig-6: Wazuh alert generated during Registry Run Key persistence creation.

Persistence Execution Investigation

Sysmon Event ID 1 — Autorun Execution Analysis

Field	Value
Event ID	1
Process	notepad.exe
Parent Process	explorer.exe
User	socadmin
Execution Type	Autorun persistence execution

Investigation Analysis

Following user logon and reboot activity, Sysmon Event ID 1 captured automatic execution of the persistence payload through a parent-child process relationship involving:

```
explorer.exe
    ↓
notepad.exe
```

This behavior strongly indicated successful autorun persistence execution because `explorer.exe` commonly launches Run Key applications during Windows user session initialization.

The automatic execution behavior confirmed that persistence had been successfully established.

Important Analyst Observation

The command line observed during execution included:

```
notepad.exe  "C:\ProgramData\Microsoft\Windows\Start  Menu\Programs\Startup\evil.txt"
```

This revealed that multiple persistence mechanisms existed simultaneously:
Registry Run Key persistence

- Startup Folder persistence

This type of multi-persistence behavior is commonly observed in real malware families to improve persistence

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Persistence Correlation Analysis

Investigation Findings

Persistence Mechanism	Type
Registry Run Key	Registry Persistence
Startup Folder	File-System Persistence

Investigation Analysis

The strongest detection indicator during this investigation was not a single event, but the behavioral correlation across multiple persistence-related telemetry sources.

The complete persistence chain consisted of:

- Registry Run Key modification
- PowerShell and reg.exe activity
- Startup folder artifact presence
- user logon autorun behavior
- explorer.exe parent-child execution chain
- successful persistence execution after reboot

This significantly increased overall detection confidence.

Important SOC Lesson

Single telemetry events are often weak indicators.

Correlated persistence activity across:

- registry modifications
- file-system artifacts
- autorun execution

reboot-triggered process creation

provides significantly higher-confidence detection opportunities during enterprise SOC investigations.

Attack Timeline

Time	Event
08:53:41	Registry Run Key created
10:33:47	Sysmon Event ID 13 generated
10:33:57	Wazuh rule alerted
09:31:54	User logon / reboot occurred
09:32:55	explorer.exe initiated autorun execution
09:32:57	Persistence payload executed

Key Detection Findings

Finding	Impact
Registry Run Key modification	High-confidence persistence indicator
reg.exe persistence creation	Suspicious registry activity
explorer.exe autorun execution	Successful persistence confirmation
Multi-persistence behavior observed	Increased attacker sophistication
Reboot-triggered payload execution	Persistence validation
Timeline correlation across telemetry	Improved detection confidence
Missing Wazuh alert generation	Detection engineering improvement opportunity

Primary Investigation Artifacts

Sysmon Event IDs

Event ID	Meaning
13	Registry Value Set
1	Process Creation
11	File Creation

Wazuh Rules

Rule ID	Description
92041	Value added to registry key has Base64-like pattern
92302	Registry entry to be executed on next logon modified using reg.exe

Important Persistence Locations

Registry Run Keys

```
HKCU\Software\Microsoft\Windows\CurrentVersion\Run
```

```
HKLM\Software\Microsoft\Windows\CurrentVersion\Run
```

Startup Folder

```
C:\ProgramData\Microsoft\Windows\Start  Menu\Programs\Startup
```

Investigation Outcome

The investigation successfully demonstrated how Sysmon provides strong visibility into Registry Run Key persistence techniques and reboot-triggered autorun execution behavior.

Behavioral correlation between:

- registry modification activity
- persistence path monitoring
- explorer.exe autorun execution
- Startup folder artifacts
- process execution telemetry

significantly improved persistence detection confidence and investigation quality.

The investigation also reinforced the importance of persistence-focused telemetry monitoring because persistence artifacts often provide high-value indicators of malicious intent and attacker access retention.

Additionally, the absence of dedicated Wazuh alerts highlighted a realistic SOC detection gap and demonstrated the importance of continuously improving SIEM correlation rules and persistence-focused detection engineering.

Conclusion

This simulation demonstrated how Sysmon provides strong visibility into

Registry Run Key persistence techniques commonly associated with malware autorun behavior and long-term attacker access retention.

Behavioral correlation between registry modification telemetry, autorun execution behavior, and persistence artifact analysis significantly improved detection confidence and investigation quality.

The investigation also highlighted the importance of multi-event timeline correlation during SOC investigations because persistence activity becomes substantially more valuable when linked to later execution behavior and reboot-triggered process activity.

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